

YOURSTORY



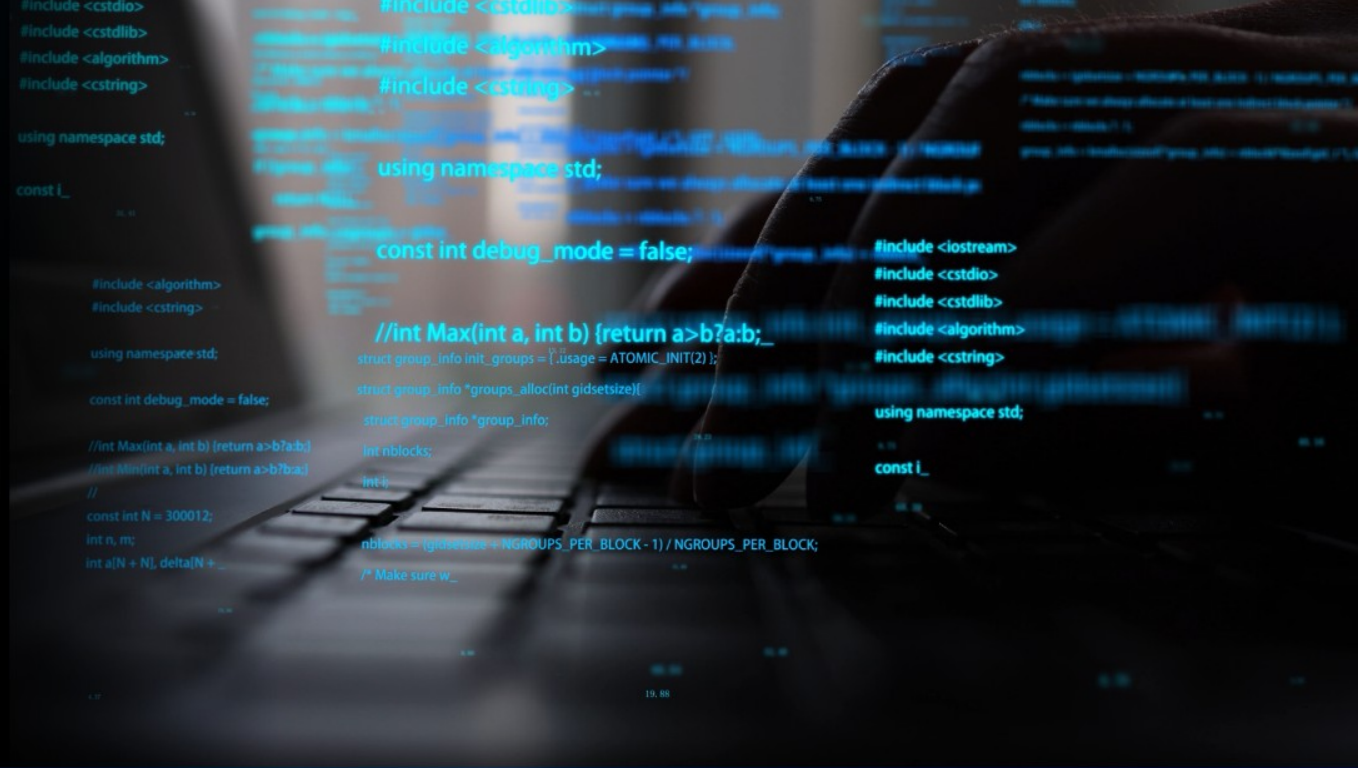
H2S



Coco CLI

HACKATHON

GCC Edition



Team Name :

Unflakeable

Project: ChainProof

Team Leader Name :

Pankaj Kumar

Team Size :

2

Problem Statement :

Supply Chain Ontology and Governed Conversational Analytics

Expanded focus: Resolve conflicting KPI definitions before AI answers

1. Problem Brief

The same KPI label can hide different business meanings - and a normal AI assistant may choose the wrong one.

Metric conflict: same label, different meaning

Planning

95%

Usable by production need date

Procurement

85%

Accepted by original PO date

Logistics

90%

Physically on time

Enterprise

85%

Approved version 1.0

Real business problem

Planning, Procurement, and Logistics use "Fill Rate" for different calculations. Valid numbers become an executive inconsistency when the label is treated as one KPI.

Target user

Primary: Supply Chain Data Steward / Operations Analytics Lead. Secondary: Planning, Procurement, Logistics, and Operations leaders.

Current pain point

Manual reconciliation, repeated meetings, supplier disputes, mixed accountability, and AI risk: correct SQL can still answer the wrong business question.

How ChainProof improves it

Detect conflict, compare contracts, assess operational impact, attach evidence, record human approval, publish only the approved version, and verify every AI answer.

Why this matters

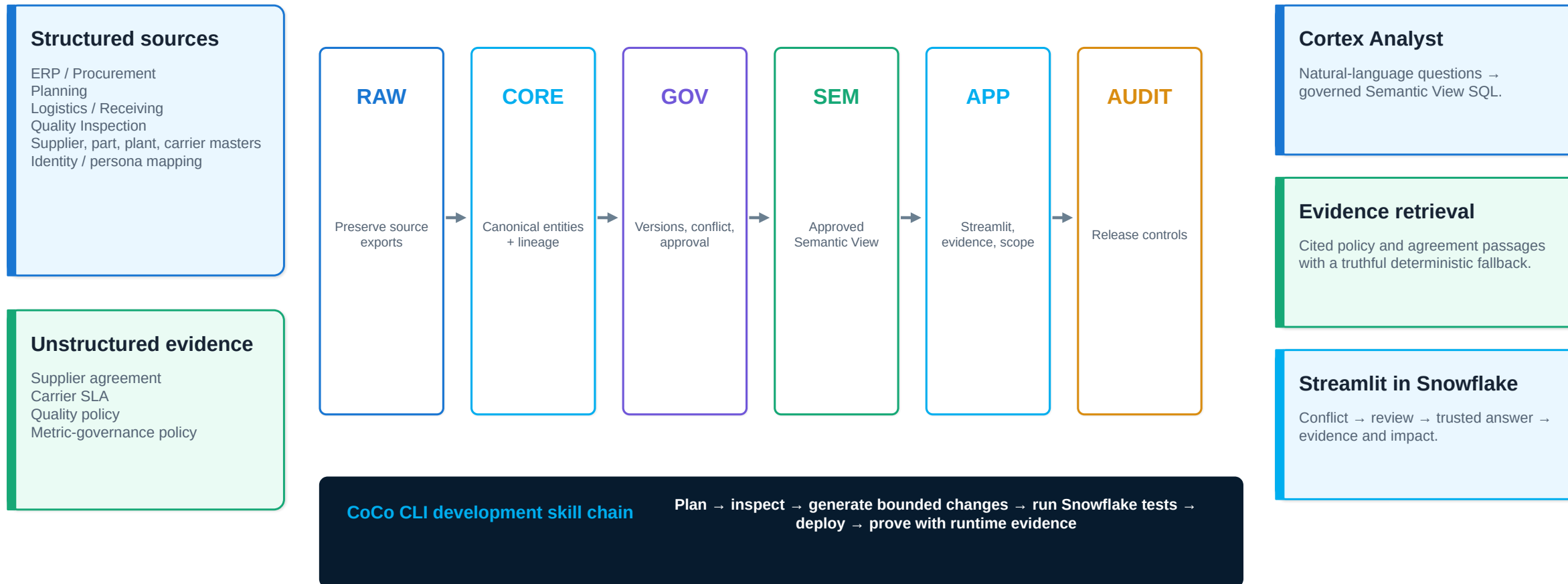
Executive consistency | supplier accountability | AI trust | faster reconciliation

Domain: laptop-manufacturing inbound supply - Supplier → Part → Purchase Order → Shipment → Plant

One KPI name. Three valid calculations. One governed enterprise answer.

2. Architecture Diagram

Snowflake-native, modular, and auditable: structured data, policy evidence, governed metrics, conversational analytics, and release controls.



Full architecture, sequence diagrams, and lifecycle proofs: [GitHub Architecture Guide](#)

Modular components plug together through governed views and immutable identifiers; each layer can evolve without rewriting the others.

CoCo CLI: Input → Processing → Output

The recorded demo shows one real read-only workflow and three modular capabilities - not only the final Streamlit screen.

1. Input - natural-language task

“Run a read-only ChainProof check for PO-5001. Query the governed conflict view, return Planning, Procurement, Logistics, and Enterprise percentages, and explain why they differ. Do not modify files or Snowflake objects.”

2. Processing - CoCo orchestrates

- Inspects the existing repository and object contract
- Uses the Snowflake CLI connection
- Executes a bounded read-only query/test
- Checks output against governed expectations

3. Output - proof, not a claim

Planning 95%
Procurement 85%
Logistics 90%
Enterprise v1.0 85%

Result is then shown in the deployed product with version, scope, and evidence.

Repository skill

Plan, inspect, generate, and review bounded changes

Snowflake execution skill

Deploy layered SQL and run deterministic fail-fast tests

Product/AI skill

Semantic View + Cortex Analyst + Streamlit + cited evidence

Video allocation: 35 sec CoCo CLI proof + 3 min 45 sec deployed workflow + 20 sec architecture and closing = 4 min 40 sec total.

3. Impact Statement

Measured MVP evidence is separated from expected production outcomes.

Measured in the MVP

- 3 competing “Fill Rate” definitions reconciled into 1 approved enterprise version
- 6 verified natural-language questions
- PO-5001: 15 acceptable units short, 10 units physically late, 5 units unavailable for production
- Scope-safe answer: PO-5001 = 85%; enterprise aggregate = 51.9%

Business outcomes

- Faster cross-functional metric alignment
- Clear supplier, carrier, and production accountability
- Visible metric scope, version, formula, dates, and evidence
- Prevention of unapproved definitions becoming trusted AI answers
- Reviewer-ready release controls and known-limitations register

Scalability and beyond the demo

- More suppliers, parts, plants, regions, and business units
- Additional metrics: OTIF, forecast accuracy, landed cost, inventory availability, supplier quality
- Live ERP/TMS/WMS/MES connectors
- Dedicated RBAC, secure approval write-back, row policies, monitoring, and audit

Impact: consistent executive reporting, faster reconciliation, stronger supplier accountability, and auditable AI answers.

Prototype / MVP Brief

A complete working workflow: source data → governed metric → conversational answer → visible evidence.

ChainProof ingests supply-chain facts and policy evidence, reconciles competing metric definitions, records human approval and version history, publishes only approved metrics through a native Semantic View, and delivers governed conversational analytics in Streamlit in Snowflake.

What works today

- Structured and unstructured source inputs
- RAW → CORE → GOVERNANCE → SEMANTIC → APP → AUDIT
- Four approved metric identities
- Enterprise version 1.0 activation history
- Native Semantic View and verified queries
- Cortex Analyst with scope guard
- Streamlit conflict, evidence, and impact experience

Fully working demo outcome

Planning 95%
Procurement 85%
Logistics 90%
Enterprise 85%

The ambiguous question “What is fill rate for PO-5001?” resolves to Enterprise Supplier Fill Rate version 1.0 = 85%, with calculation and policy evidence.

Production path

- Replace CSVs with live ERP/TMS/WMS/MES feeds
- Dedicated owner/viewer/Data Steward roles
- Secure approval procedure and audit trail
- Broader metric catalog and effective dating
- Search/Agent where account capabilities permit
- CI/CD, monitoring, cost controls, and access policies

[Public repository](#) | [Live application](#) | [Walkthrough video](#)



ChainProof

Metric Trust and Reconciliation for Supply-Chain AI

Snowflake-native

CoCo CLI-assisted

One KPI name. Three valid calculations. One governed enterprise answer.

Planning

Material Availability

95%

Procurement

Accepted Supplier Fill

85%

Logistics

On-Time Arrival

90%



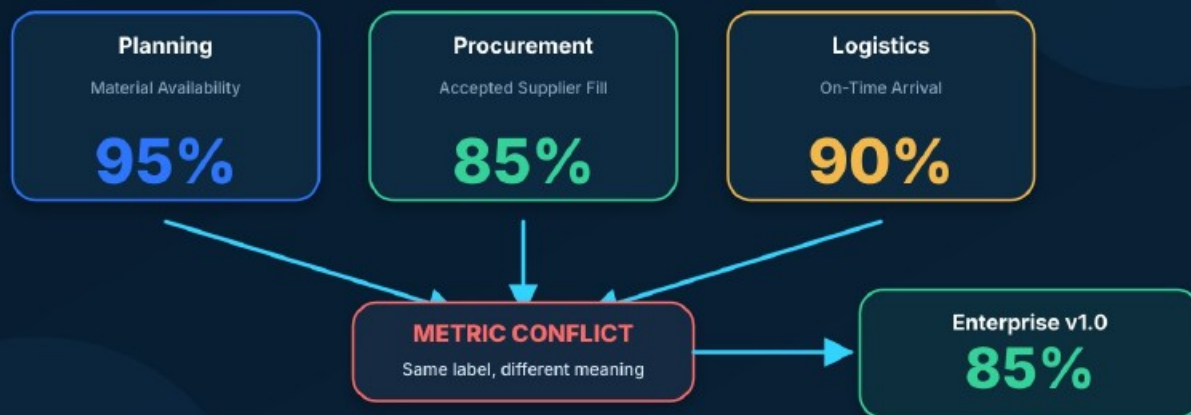
**Enterprise Supplier Fill Rate v1.0 =
85%**

The business problem

Correct data and valid SQL can still produce the wrong business answer.

One KPI name. Three valid calculations.

ChainProof governs the disagreement before AI answers.



Detect → Compare → Assess impact → Evidence → Approve → Publish → Verify

Executive inconsistency

Different teams publish incompatible KPIs under one label.

Supplier disputes

Date and quality rules change accountability.

AI risk

A copilot can confidently select the wrong definition.

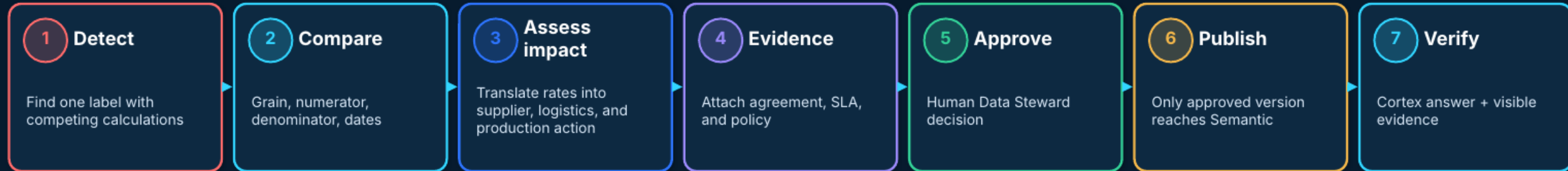
Manual reconciliation

Analysts repeatedly rebuild the same explanation.

Correct SQL + wrong metric meaning = confidently wrong AI

The ChainProof trust lifecycle

AI analytics begins only after the metric is governed.



Metric trust firewall

A normal copilot answers the question. ChainProof first proves which definition it is allowed to use.

USP: Semantic layers govern agreed metrics. ChainProof governs the disagreement that exists before agreement.

Human governance changes the answer

The AI consumes the Data Steward decision; it does not make it.

BEFORE APPROVAL

Planning	95%
Procurement	85%
Logistics	90%

Enterprise answer: NOT APPROVED

AFTER DATA STEWARD APPROVAL

Enterprise Supplier Fill Rate

85%

Version 1.0 | Enterprise - Approved

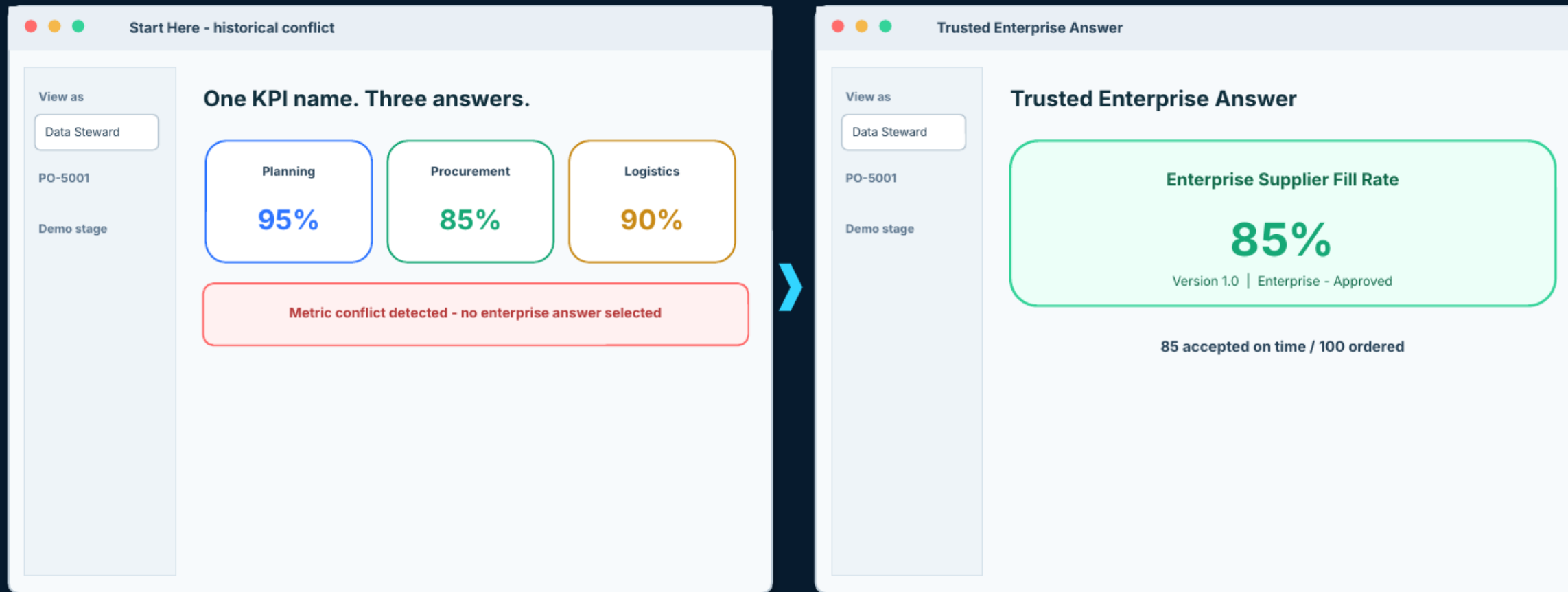
85 accepted on time / 100 ordered

Approved → Active → Published

“View as” demonstrates persona presentation under one learner role. It never changes permissions or formulas.

Judge-first product journey

The deployed app tells the business story before showing architecture.



Start with the bad state

End with the governed answer

Snowflake-native architecture

Each layer has one responsibility and a deterministic validation gate.

Snowflake-native trust architecture

Operational facts, metric governance, AI analytics, and audit remain separated.



Core rule
Only an approved metric version may become a trusted AI answer.

- Semantic View** Only approved metrics are public
- Cortex Analyst** Natural-language governed SQL
- Streamlit in Snowflake** No external app credential required
- Evidence workflow** Citations + trusted boundary
- AUDIT** Release controls + limitations

RAW → CORE → GOVERNANCE → SEMANTIC → APP → AUDIT [Full architecture guide](#)

Trust controls around every AI answer

ChainProof does not trust generated SQL merely because it compiles.

1

Question scope

PO-specific vs enterprise aggregate

2

Read-only SQL

One SELECT/WITH statement only

3

Semantic boundary

No RAW, CORE, or GOVERNANCE access

4

Version passport

Name, scope, owner, version, formula

5

Evidence trust

Citations; injection fixture excluded

6

Human authority

Advisor cannot approve or write

Fail visibly. Preserve the approved formula. Never fabricate a capability or answer.

Evidence and operational impact

The governed number is explainable and connected to a concrete business action.

View as

Data Steward

PO-5001

Demo stage

Evidence-backed review

Supplier agreement [cited]

Carrier SLA [cited]

Quality policy [cited]

Governance policy [cited]

Advisor cannot approve or write governance data

PO-5001 business impact

Enterprise Supplier Fill Rate

85%

against 90% threshold

15 units

supplier shortfall

Procurement

15 acceptable units short by original PO date

Logistics

10 units arrived after carrier commitment

Planning

5 usable batteries short; up to 5 laptops at risk

Action: supplier escalation + carrier follow-up + production mitigation

One governed metric, three clearly separated operational consequences.

Technical execution and validation

A twelve-part build with deterministic gates, real account evidence, and truthful limitations.

12

controlled build parts

6

Snowflake schemas

4

approved public metrics

6

verified questions

1

deployed Streamlit app

RAW

files, rows, edge cases

CORE

types, lineage, quality

GOVERNANCE

versions, approval, activation

SEMANTIC

metrics, relationships, verified SQL

APP

scope, persona, evidence, SQL safety

AUDIT

release controls and limitations

Known limitation: official Analyst batch evaluation needs account-level task privileges. Live Cortex Analyst and deterministic tests passed.

Why ChainProof is different

It governs metric disagreement - the gap before a normal semantic layer can help.

Category	Typical capability	ChainProof addition
BI dashboard	Displays configured KPIs	Detects conflicting KPI meaning
Data catalog	Documents names and owners	Executes formulas and compares impact
Semantic layer	Publishes agreed definitions	Governs disagreement before agreement
Generic AI copilot	Answers with available metadata	Refuses ambiguity until governed
Metric store	Centralizes formulas	Preserves conflict, approval, activation, rollback

Business value: fewer conflicting reports, faster reconciliation, stronger supplier accountability, and auditable AI answers.

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GCC Edition

THANK YOU

Team Unflakeable | ChainProof

One KPI name. Three valid calculations. One governed enterprise answer.

[GitHub](#) | [Live app](#) | [Video](#)